

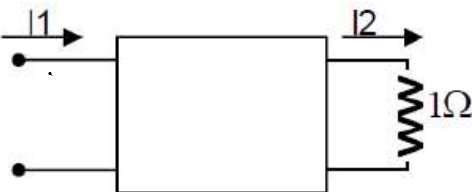
Ejercicio 8 - Hoja 1

sábado, 13 de septiembre de 2025

21:23

8) Sintetice el siguiente cuadripolo RC:

$$-\frac{I_2}{I_1} = H \cdot \frac{(s^2 + 5s + 4)}{(s^2 + 8s + 12)}$$

$$Z_{21} = 6H$$


• Comenzamos igualando la expresión $-\frac{I_2}{I_1}$ a la del cuadripolo

$$-\frac{I_2}{I_1} = \frac{1}{A \cdot Y_g \cdot Z_L + C \cdot Z_L + Y_g \cdot D + D}, \text{ donde por simple inspección queda } Y_g = 0, Z_L = 1$$

$$-\frac{I_2}{I_1} = \frac{1}{C \cdot Z_L + D}, \quad C = \frac{1}{Z_{21}}, \quad D = \frac{Z_{22}}{Z_{21}}$$

$$-\frac{I_2}{I_1} = \frac{1}{\frac{1}{Z_{21}} \cdot Z_L + \frac{Z_{22}}{Z_{21}}} \cdot \frac{Z_{21}}{Z_{21}} = \frac{Z_{21}}{Z_L + Z_{22}} = \frac{Z_{21}}{1 + Z_{22}}$$

$$\boxed{-\frac{I_2}{I_1} = \frac{Z_{21}}{1 + Z_{22}}} \Rightarrow \text{Igualando} \Rightarrow \frac{Z_{21}}{1 + Z_{22}} = H \cdot \frac{(s^2 + 5s + 4)}{(s^2 + 8s + 12)}$$

$$\Rightarrow \frac{6H}{1 + Z_{22}} = H \cdot \frac{(s^2 + 5s + 4)}{(s^2 + 8s + 12)} \Rightarrow 6 \cdot \frac{(s^2 + 8s + 12)}{(s^2 + 5s + 4)} = (1 + Z_{22})$$

$$\Rightarrow 6 \cdot \frac{(s^2 + 8s + 12)}{s^2 + 5s + 4} = 1 + Z_{22}$$

$$\Rightarrow \frac{6(s^2 + 8s + 12)}{s^2 + 5s + 4} - 1 = Z_{22}$$

$$\Rightarrow \frac{6s^2 + 48s + 72 - s^2 - 5s - 4}{s^2 + 5s + 4} = Z_{22}$$

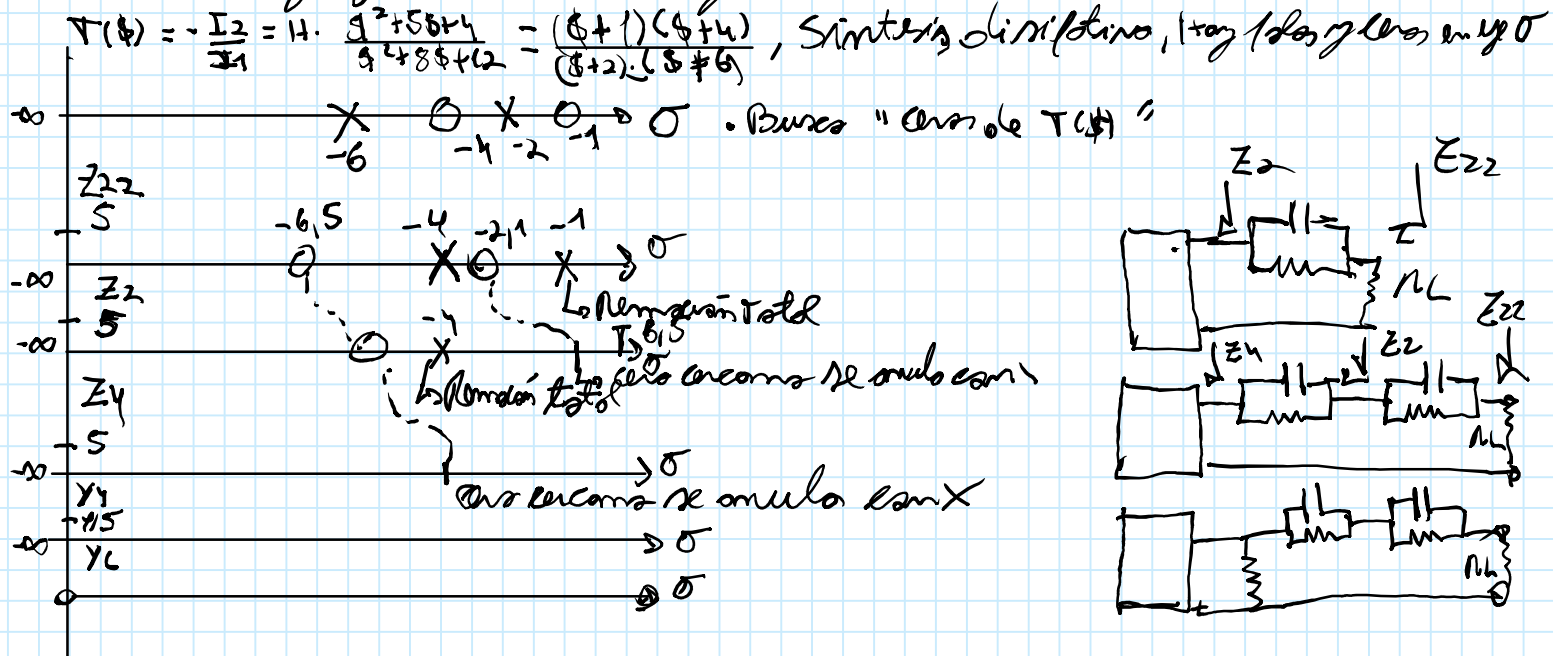
$$\Rightarrow \boxed{\frac{5s^2 + 43s + 68}{s^2 + 5s + 4} = Z_{22}} \Rightarrow \frac{5(s^2 + 8.6s + 13.6)}{(s+1)(s+4)} = \frac{5(s+3.1)(s+6.5)}{(s+1)(s+4)}$$

$$\boxed{x^2 + 5x + 4}$$

$$\overline{(x+1)(x+4)} \quad (x+1)(x+4)$$

$$Z_{22}(s) = \frac{5 \cdot (s+2+1) \cdot (s+6,5)}{(s+1) \cdot (s+4)} \rightarrow \text{función de transferencia}$$

1º) Métodos gráficos de síntesis y remoción



El último elemento a remover será un R puesto que hay una $\pm j\omega$ en ∞ . En particular, esto se debe a la derivación y a la entrada del circuito puesto que el generador de entrada es de corriente.

2º) Cálculo de Residuos

$$Z_2(s) = Z_{22}(s) - \frac{2K_1 \cdot 1}{s+1}, \quad Z_2(s) \Big|_{s=-1} = 0 \Rightarrow (Z_{22}(s) - \frac{2K_1}{s+1}) \Big|_{s=-1} = 0$$

$$\frac{5(s+2+1)(s+6,5)}{(s+1)(s+4)} - \frac{2K_1}{s+1} = 0 \Rightarrow \frac{5(s+2+1)(s+6,5)}{(s+1)(s+4)} = \frac{2K_1}{s+1}$$

$$\frac{5(-1+2+1)(-1+6,5)}{(-1+4)} = 10,83 = 2K_1 \approx 10$$

$$Z_2(s) = \frac{5(s+2+1)(s+6,5)}{(s+1)(s+4)} - \frac{10}{s+1} = \frac{5(s+2+1)(s+6,5) - 10(s+4)}{(s+1)(s+4)}$$

$$Z_2(s) = \frac{5s^2 + 43s + 68 - 10s - 40}{(s+1)(s+4)} = \frac{5s^2 + 33s + 28}{(s+1)(s+4)} = \frac{5(s+1)(s+5,6)}{(s+1)(s+4)}$$

Ejercicio 8 - Hoja 3

sábado, 13 de septiembre de 2025

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$$Z_2(s) = \frac{5 \cdot (s+5.6)}{(s+4)}, \quad Z_2(0) = \frac{5(0+5.6)}{0+4} = 7$$

$$Z_4(s) = Z_2(s) - \frac{2K_2}{s+4}, \quad Z_4(s) \Big|_{s=-4} = 0 = \left(Z_2(s) - \frac{2K_2}{(s+4)} \right) \Big|_{s=-4}$$

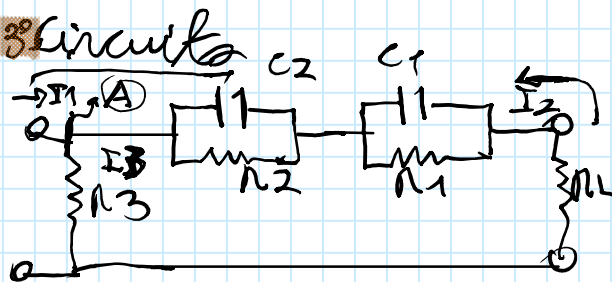
$$\frac{5 \cdot (s+5.6)}{(s+4)} - \frac{2K_2}{(s+4)} = 0 \Rightarrow \frac{5(s+5.6)}{(s+4)} = \frac{2K_2}{(s+4)} \Rightarrow 5 \cdot (s+5.6) = 2K_2$$

$$\Rightarrow 5(-4+5.6) = 2K_2 \Rightarrow 2K_2 = 8$$

$$Z_4(s) = Z_2(s) - \frac{8}{s+4} = \frac{5(s+5.6)}{(s+4)} - \frac{8}{(s+4)} = \frac{5s+28-8}{s+4} = \frac{5s+20}{s+4} = 5$$

$$Z_4(s) = 5, \quad Y_4(s) = \frac{1}{5}$$

$$Y_6(s) = Y_4(s) - \frac{1}{5} = 0$$



$$Z_{C1R1} = \frac{2K}{s+0}$$

$$Z_{C1R1} = \frac{1}{Y_{C1R1}} = \frac{1}{\frac{1}{s} + \frac{1}{R1}} \cdot \frac{1/C1}{1/C1} = \frac{1/C1}{s + \frac{1}{R1C1}}$$

$$\Rightarrow Z_{C1R1} = \frac{1/C1}{s + \frac{1}{C1R1}} \Rightarrow \begin{cases} 1/C1 = 2K1 = 11 \\ C1 = \frac{1}{10} F \\ \frac{1}{C1R1} = 1 \\ R1 = 0.1 \end{cases}$$

$$\Rightarrow Z_{C2R2} = \frac{1/C2}{s + \frac{1}{C2R2}} \Rightarrow \begin{cases} \frac{1}{C2} = 2K2 = 7 \\ C2 = \frac{1}{8} F \\ \frac{1}{C2R2} = 4 \\ R2 = 0.2 = 2 \Omega \end{cases}$$

$$\Rightarrow R3 = 3$$

4º Verificación

Observamos el circuito, aplicamos Karchoff ley de mallas:

$$V_{R3} = V_{R1} + V_{C1R1} + V_{R2C2} \Rightarrow \text{malla (A)} \quad I_1 + (-I_2) = I_3, \quad Z' = Z_{C1R1} + Z_{C2R2}$$

$$Z' = \frac{1/C1}{s + \frac{1}{C1R1}} + \frac{1/C2}{s + \frac{1}{C2R2}} = \frac{11}{s+1} + \frac{7}{s+4} = \frac{11(s+4) + 7(s+1)}{(s+1)(s+4)} = \frac{11s+44+7s+7}{s^2+5s+4}$$

$$Z = \frac{1}{s+1} + \frac{1}{s+2} + \frac{1}{s+1} + \frac{1}{s+4} = \frac{(s+1)(s+4) + (s+1)(s+2) + (s+1)(s+4) + (s+1)(s+2)}{(s+1)(s+4)}$$

$$Z' = \frac{18s+51}{s^2+5s+4} = 18 \frac{s+51}{s^2+5s+4}, \quad (I_1 + I_2) \cdot s^3 = I_2 \cdot Z' + I_2 \cdot n_2$$

$$I_1 \cdot n_3 = I_2 \cdot Z' + I_2 \cdot n_2 + I_2 \cdot n_3$$

Ejercicio 8 - Hoja 4

domingo, 14 de septiembre de 2025 00:28

$$(I_1 + I_2)R_3 = I_2 \cdot Z' + I_2 \cdot R_1 \Rightarrow I_1 \cdot R_3 = I_2 \cdot Z' + I_2 \cdot R_1 + I_2 \cdot R_3$$

$$\Rightarrow \frac{R_3}{Z' + R_1 + R_3} = \frac{I_2}{I_1}$$

$$\Rightarrow \frac{5}{18\left(5 + \frac{51}{18}\right) + 1 + 5} = \frac{(x^2 + 5x + 4)}{x^2 + 5x + 4} = \frac{I_2}{I_1}$$

$$\Rightarrow \frac{5 \cdot (x^2 + 5x + 4)}{18x + 51 + 6(x^2 + 5x + 4)} = \frac{5(x^2 + 5x + 4)}{6x^2 + 48x + 75} = \frac{5(x^2 + 5x + 4)}{6(x^2 + 8x + 12)} = \frac{I_2}{I_1}$$

• Que lo verificamos y se obtiene que $x = 5/6$